Predictive Approaches to Treatment Effect Heterogeneity

Tom Delany, Misgana Ojamo[1]

[1] School of Mathematical and Statistical Sciences, University of Galway

Ollscoil na Gaillimhe University of Galway

Background

- Randomised controlled trials are typically reported in terms of average treatment effects. This assumes homogenous patient response and ignores patients differences in demographics, disease severity, and other contextual factors - termed patient **Heterogeneity (HTE)** [1], crucially, these factors may modify the effectiveness of treatment.

- Traditional **subgroup analyses** attempt to address HTE by examining one variable at a time (e.g Male vs Female). However, these analyses are frequently underpowered, suffer from high rates of false positives and miss multivariate treatment response patterns.

- **Predictive Approaches to Treatment effect Heterogeneity (PATH)** shift the focus to individualized prediction, modelling risk and treatment effects across individuals [2].

Objectives:

1. Implement and evaluate predictive HTE models across multiple clinical datasets.
2. Assess the clinical relevance and reporting implications of PATH analyses.

Our Trials

Analyses are based on published clinical trial data from the GUSTO-I trial (GUSTO) [3] and the Digitalis Investigation Group trial (DIG) [4], the characteristics of which are described below.

Overview of GUSTO and DIG trials

Characteristic	GUSTO	DIG
Patients (n)	41,021	6,800
Treatments	Accelerated tPA vs Streptokinase (± heparin)	Digoxin vs Placebo
Years ran	1989–1993	1991–1995
Design	Open-Label	Double-blind, placebo-controlled
Follow-up	30 days	37 Months (mean)
Location	International - 15 Countries	USA - 302 Centres
Primary outcome	30-day mortality	All-cause mortality

Early Results

DIG	
Variable	**N = 6,800**[1]
AGE	65 (57, 71)
EJF_PER	29 (22, 35)
Classification	
NYHA_1	907 (13%)
NYHA_2	3,664 (54%)
NYHA_3	2,081 (31%)
NYHA_4	142 (2.1%)

[1] Median (Q1, Q3); n (%)

GUSTO	
Variable	**N = 29,448**[1]
Age (years)	62 (52, 70)
MI location	
Inferior	17,582 (60%)
Anterior	11,866 (40%)
Sex	
male	22,015 (75%)
female	7,433 (25%)

[1] Median (Q1, Q3); n (%)

Figure 1: Key subgroups and their characteristics for DIG and GUSTO. MI - Myocardial Infarction, NYHA - New York Heart Association, EJF_PER - Ejection fraction (percent)

PATH Compatible Forest Plots

To evaluate how our effects stratified by baseline risk compare with effects observed in traditional subgroups, we generated the following forest plots for the DIG and GUSTO trials.

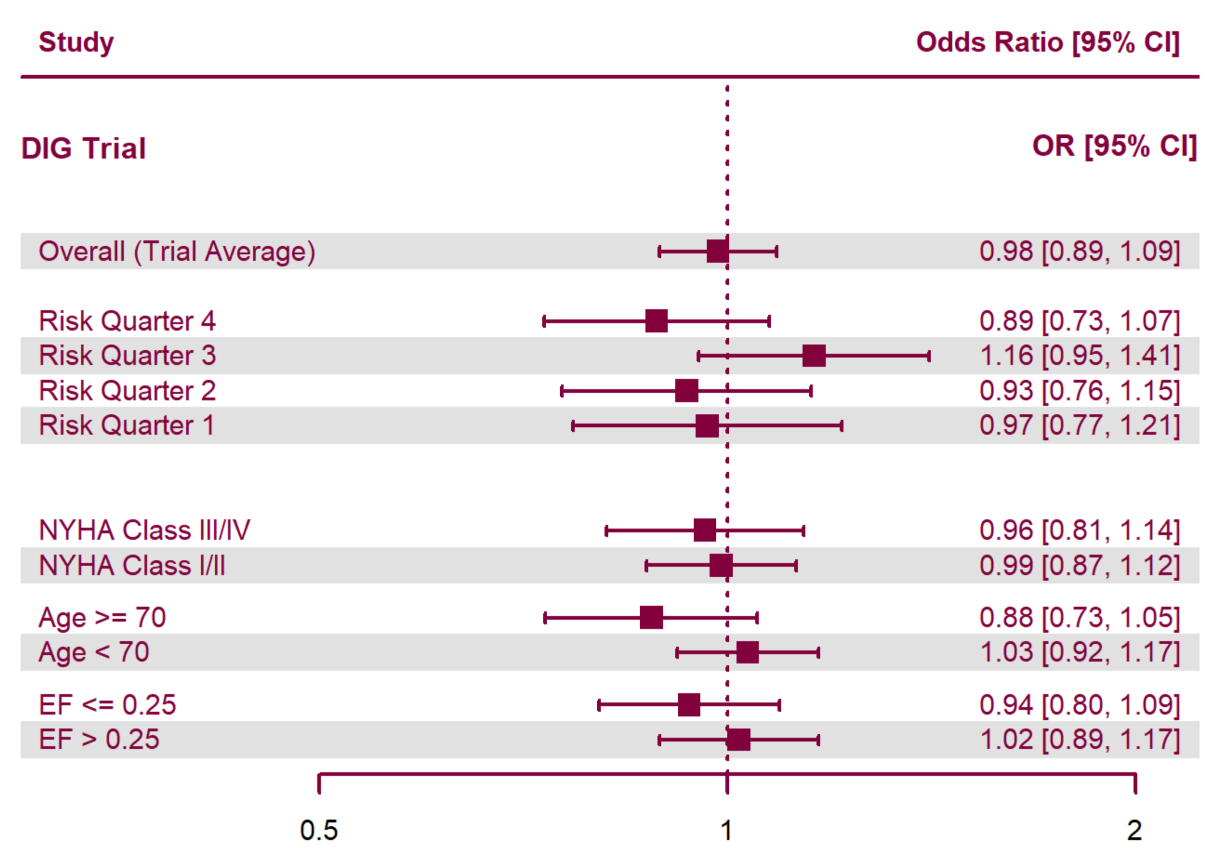

Figure 2: DIG Trial, Forest plot of treatment heterogeneity. Points represent odds ratios and horizontal lines indicate 95% confidence intervals.

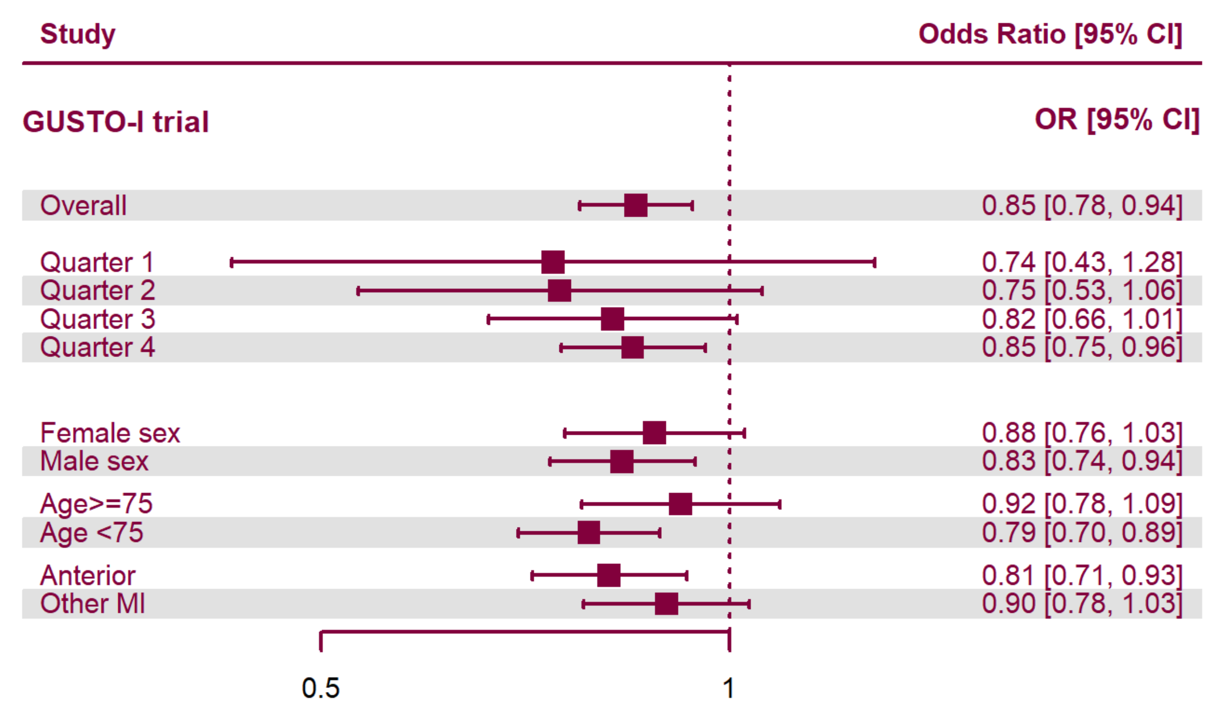

Figure 3: GUSTO Trial. Forest plot of treatment heterogeneity. Points represent odds ratios and horizontal lines indicate 95% confidence intervals; the vertical line denotes no effect (OR = 1).

Absolute Benefit of Treatment by Baseline Risk

- Absolute treatment benefit from a main-effects model **increases with baseline risk;** the solid maroon curve shows expected risk reduction under a proportional treatment effect assumption.

- Proportional-effect and **spline-smoothed** (dashed red curve) approaches yield similar estimates of treatment benefit, indicating increasing absolute benefit with higher baseline risk.

- The histogram at the bottom of the plot shows substantial **heterogeneity in baseline risk**, concentrated in low risk values.

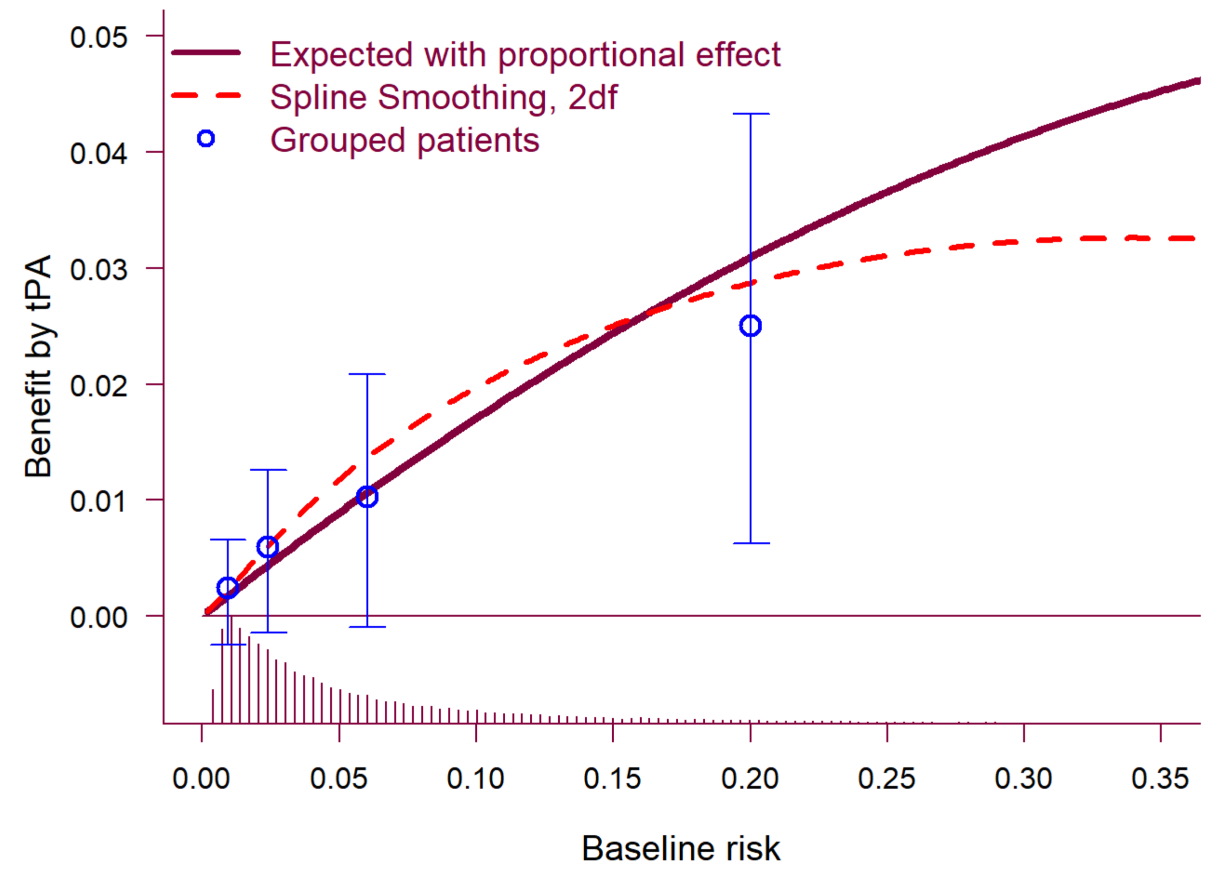

Figure 4: Absolute treatment benefit increases with baseline risk across estimates in the GUSTO trial.

Next Steps

We will conduct further analyses on the International Stroke Trial (IST) [5] and the National Lung screening Trial (NLST) [6], to evaluate the applicability of PATH across domains. We also plan to implement a machine-learning approach using causal forests for patient risk prediction. We will use the `grf` package for this.

Overview of IST and NLST trials

Characteristic	IST	NLST
Patients (n)	19,435	53,254
Treatments	Factorial (± Heparin ± Aspirin)	CT screening vs. chest radiography
Years ran	1991–1996	2002-2009
Design	Open-Label	Randomized controlled trial
Follow-up	14 days (early outcomes); 6 months (main)	Median 6.5 years
Location	International - 36 Countries	USA - 33 centres
Primary outcome	14-day and 6 month mortality	Lung-cancer mortality

GitHub

The code and datasets for this project can be viewed at our GitHub repository here: https://github.com/tomdelany04/PATH-Master/

References

1. Kent et al. 2020, doi: 10.7326/M18-3668↩
2. Willke et al. 2012, doi: 10.1186/1471-2288-12-185↩
3. The GUSTO investigators, 1993, doi: 10.1056/NEJM199309023291001↩
4. The Digitalis Investigation Group, Feb. 1997, doi: 10.1056/NEJM199702203360801↩
5. Sandercock et al., 2011, doi: 10.1186/1745-6215-12-101↩
6. The National Lung Screening Trial Research Team, 2011, doi: 10.1056/NEJMoa1102873↩